Online Resource 1

Supplementary material for the article "AXIOM: A Shared Controller for Scalable Multi-Agent 5G Slice Admission"

Wireless Personal Communications

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S1. Overview

This supplement gives the detail behind the main paper's results for AXIOM, short for Adaptive Cross-cell Integrated Orchestration of Multi-slice admission, without repeating what the main paper states. Every pass criterion was fixed in a pre-registration committed before the run it judged, and Table S1 lists the acronyms used here.

Table S1 Acronyms used in this supplement

Acronym	Meaning	Acronym	Meaning	Acronym	Meaning
CI	Confidence interval	gNB	gNodeB, the 5G base station	RF	Radio frequency
CPU	Central processing unit	GPU	Graphics processing unit	RIC	RAN Intelligent Controller
DP	Differential privacy	IQM	Interquartile mean	SDR	Software-defined radio
DP-SGD	Differentially private stochastic gradient descent	MAPPO	Multi-agent proximal policy optimisation	SLA	Service Level Agreement
DQN	Deep Q-network	MLP	Multi-layer perceptron	TDD	Time-division duplex
DQN-I	Independent per-gNB deep Q-networks	mMTC	Massive Machine-Type Communication	UDP	User Datagram Protocol
DQN-SA	Single-agent deep Q-network	OAI	OpenAirInterface	UE	User equipment
eMBB	Enhanced Mobile Broadband	O-RAN	Open Radio Access Network	URLLC	Ultra-Reliable Low-Latency Communication
FedAvg	Federated averaging	PRB	Physical Resource Block	VDN	Value-decomposition network
GAT	Graph attention network	RAN	Radio Access Network		

S2. Verification runs

The fresh-seed runs of the main paper's Section 6.6 retrained every policy on seeds from 3000 onwards. In Env A AXIOM reached 98.2% of the static rule's reward at three gNBs and at nineteen gNBs kept 95.6% priority-slice compliance against 47.8% to 65.5% for the baselines. Of the nine failed checks, the four that concern AXIOM are small margins, and twenty-seed reruns did not reproduce two of them.

Fig. S1 sets each reported comparison with a learned baseline beside its fresh-seed counterpart. The five reported differences that fresh seeds do not keep are secondary. Two are AXIOM's leads over DQN-SA in Env A at three gNBs, 0.104 reward per gNB per step and 1.9 points of priority-slice compliance. Both became clear only with the added seeds and fall to 0.033 and 0.4 on fresh seeds. The other three concern cluster-level compliance in Env B. AXIOM's deficits of 9.2 points to DQN-SA and 4.8 points to DQN-I at seven gNBs become 4.2 points and a 0.8-point lead. Its lead over MAPPO at nineteen gNBs falls to 2.9 points, and all five fresh CIs span zero.

Online Resource 2, a CSV file, gives the interquartile means and probabilities of improvement that Agarwal et al., cited in the main paper, recommend for evaluations with few runs, computed with rliable 1.2.0 on the main paper's pooled runs. For each setting and policy it lists the interquartile means of reward per gNB per step and of priority-slice compliance. It also gives the probability that a run of AXIOM beats a run of the other policy, with ties counted as one half. Each value has a stratified bootstrap 95% CI over 10,000 resamples. The IQM drops the worst quarter of runs, where AXIOM's few collapses sit. AXIOM's IQM of priority-slice compliance in Env A is therefore 100.0% at every size, against means of 94.9% to 99.0%, and the mean stays the more conservative summary for AXIOM. An independent implementation reproduces every IQM and every probability of improvement to within 10^{-6}, and Fig. S2 shows the fraction of runs above each compliance level.

Fig. S3 sets the same interquartile means and probabilities of improvement side by side for every setting.

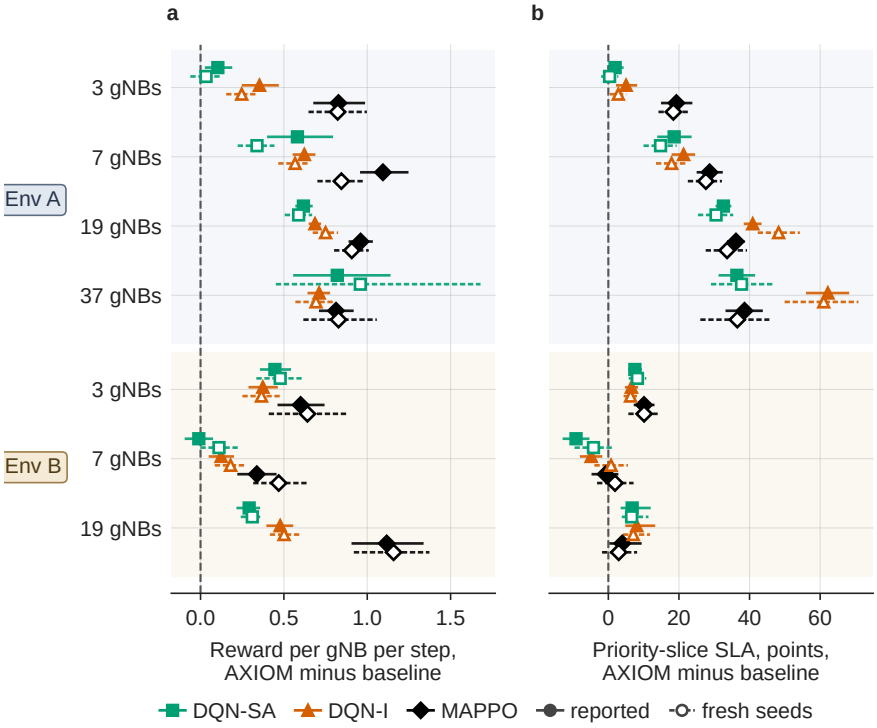


Fig. S1 AXIOM minus each learned baseline, with 95% CIs. Solid lines and filled circles are the reported runs, dashed lines and hollow diamonds the fresh seeds. a, Reward per gNB. b, Cluster-level priority-slice compliance

S3. Live testbed

Every live measurement comes from the single-host, software-radio O-RAN stack of Fig. S4, which runs without a near-real-time RIC because the ORANSlice fork cited in the main paper applies slicing control through a lightweight UDP E2 agent. It is the stack on which our recently presented conference paper, cited in the main paper and due to be published in IEEE Xplore, deployed its single-gNB DQN admission controller. AXIOM itself is not deployed there, since its multi-gNB policy needs at least two live gNBs and our two-gNB attempt did not sustain traffic. A single-agent DQN checkpoint instead exercises the control path, PRB-ratio actuation, and real-time inference at one and at two UEs per slice. A capacity probe showed every slice served at about 100% of offered demand, so the rig never ran short of capacity and shows the pipeline working, not admission under contention. The rig starts the Open5GS core with Docker Compose and the gNB with `nr-softmodem` on the rfsim radio, and attaches up to six UEs. It runs the checkpoint through the UDP E2 agent on ports 6600 and 6655. It drives iperf3 UDP traffic on every slice to load the rig. A scheduler-configuration error that granted fewer PRBs than the gNB scheduler's five-PRB floor was fixed at the root. Instrumentation then confirmed zero below-floor grants across more than 100,000 readings.

S4. Setup, safeguards, and analysis plans

A sensitivity study repeats the learned policies and the static rule in Env A on 10 seeds at 102% and 107% of the nominal load. It also changes the slice weights, lowering eMBB from 3.5 to 2.5 and raising mMTC from 0.3 to 0.6. A second repeats AXIOM in Env B on 10 seeds at learning rates of $3{\times}10^{-4}$ and $3{\times}10^{-3}$ and with an encoder twice as wide. The federated campaign runs in Env A on seeds 900 to 909, with noise multipliers $\sigma \in \{0, 0.5, 1, 2, 4\}$ and a replication on seeds 1000 to 1009. Each round runs 50 local steps of federated averaging (FedAvg) with a gradient-clip norm of 1.0. A validation screen checks every trained checkpoint before evaluation. Each plays five greedy episodes on its own training stream and fails if it blocks no request or if fewer than 90% of its blocks target mMTC. In Env B, where the reward-optimal policy also sheds eMBB, the no-blocking criterion is the primary screen. The Env B constants, AXIOM's learning rate, and its skip connection were all chosen on pilot seeds 2000 to 2009.

AXIOM's encoder is a graph-attention network over the gNB graph. Each gNB forms its own feature row $\mathbf{x}_i$ of PRB-used ratio, congestion level, and normalised queue length, and a two-layer network with $H{=}4$ heads produces one embedding per gNB. Each head h uses a learned source vector $\mathbf{a}_s^h$ and destination vector $\mathbf{a}_d^h$ on the shared projection $\mathbf{W}^h$. Here LeakyReLU, the leaky rectified linear unit, has slope 0.2 for negative inputs, and $\mathcal{N}_i$ is the set of neighbours of node i,

$$\varepsilon_{ij}^h = \mathrm{LeakyReLU}\big(\mathbf{a}_s^{h\top}\mathbf{W}^h\mathbf{x}_i + \mathbf{a}_d^{h\top}\mathbf{W}^h\mathbf{x}_j\big), \qquad \alpha_{ij}^h = \frac{\exp\varepsilon_{ij}^h}{\sum_{l\in\mathcal{N}_i}\exp\varepsilon_{il}^h}. \tag{1}$$

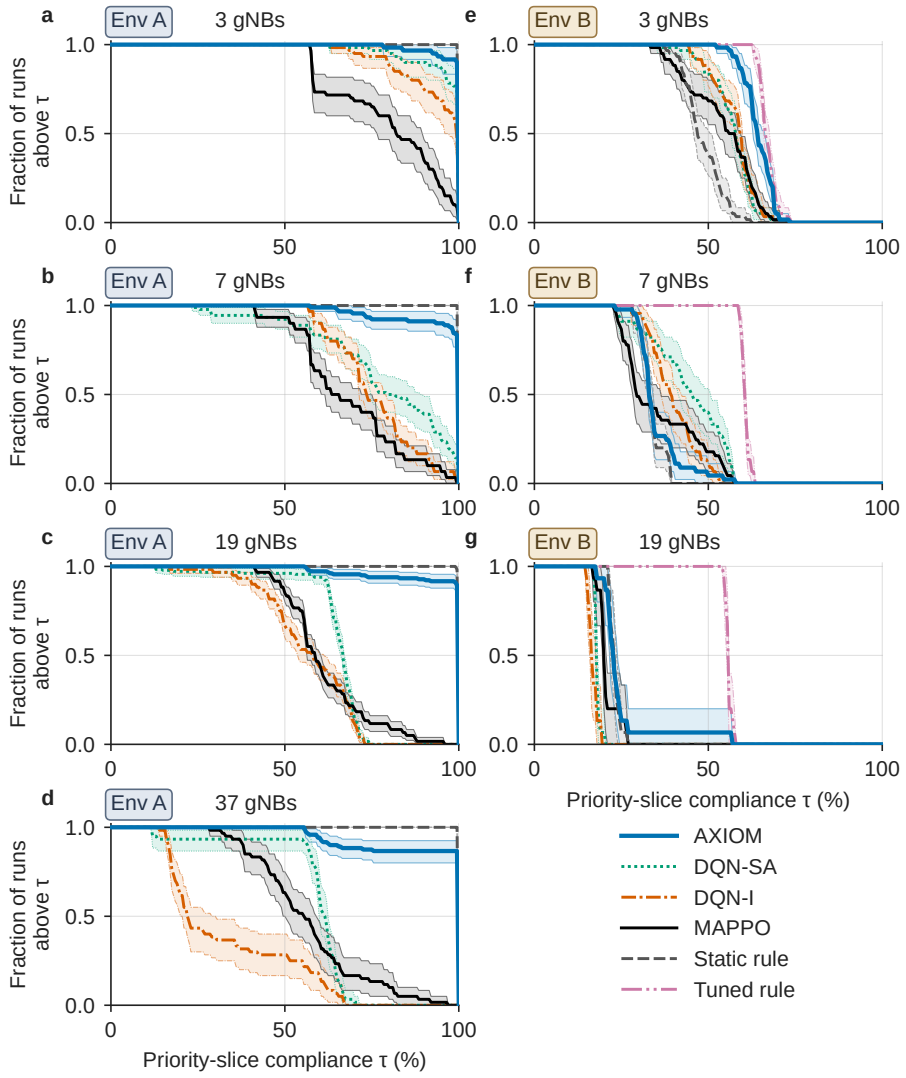


Fig. S2 Performance profiles of priority-slice compliance, the fraction of runs above each level τ. Bands bounded by thin lines are stratified bootstrap 95% CIs. a to d, Env A at 3, 7, 19, and 37 gNBs. e to g, Env B at 3, 7, and 19 gNBs

Hidden layers concatenate the H heads under an exponential linear unit, the final layer averages them, and a per-layer normalisation $\mathrm{LN}(\cdot)$ rescales each embedding to zero mean and unit variance,

$$\mathbf{z}_i = \mathrm{LN}\Big(\tfrac{1}{H}\textstyle\sum_{h=1}^{H}\sum_{j\in\mathcal{N}_i}\alpha_{ij}^{h}\,\mathbf{W}^{h}\mathbf{x}_j\Big). \tag{2}$$

An earlier variant fed each gNB's summed per-request values into a QMIX-style mixer, and its mean training loss diverged from about 76 to about 409. Treating each request as its own sample removed that dependence.

In the federated variant each client clips its minibatch gradient to norm C_{dp} and adds Gaussian noise of standard deviation σC_{dp}. Zero-concentrated differential privacy composes this over the m noisy steps a client takes,

$$\rho = \frac{2m}{\sigma^2}, \qquad \varepsilon(\delta) = \rho + 2\sqrt{\rho\ln(1/\delta)}. \tag{3}$$

We credit no amplification by subsampling, because minibatches come from a replay buffer that reuses records, so the bound is conservative.

The trivial-policy check of the main paper's Section 5 rests on these numbers. In our first simulator, where every slice was served against its own ceiling, accepting every request earned 14.45 reward per step and kept URLLC and eMBB at 100% SLA compliance. Over the same 30 held-out seeds in the shared-budget simulator, accepting every request now earns 9.89 reward per step and lets eMBB compliance fall to 56.8%, whereas shedding every mMTC request earns 12.84.

The main campaign reuses Env A results of DQN-SA, DQN-I, and the static rule from an earlier rerun, M63. They cover three gNBs on the primary and replication seeds, the sensitivity study, seven gNBs, and 15 of the 30 seeds at nineteen gNBs. Two development stages are disclosed, although neither changed a reported result. A protocol, M67, tested checkpoint selection and both encoders on new seeds and left AXIOM unchanged, and exploratory probes on seeds 900 to 904 suggested the load-shift effect of Section S7 before its reported runs.

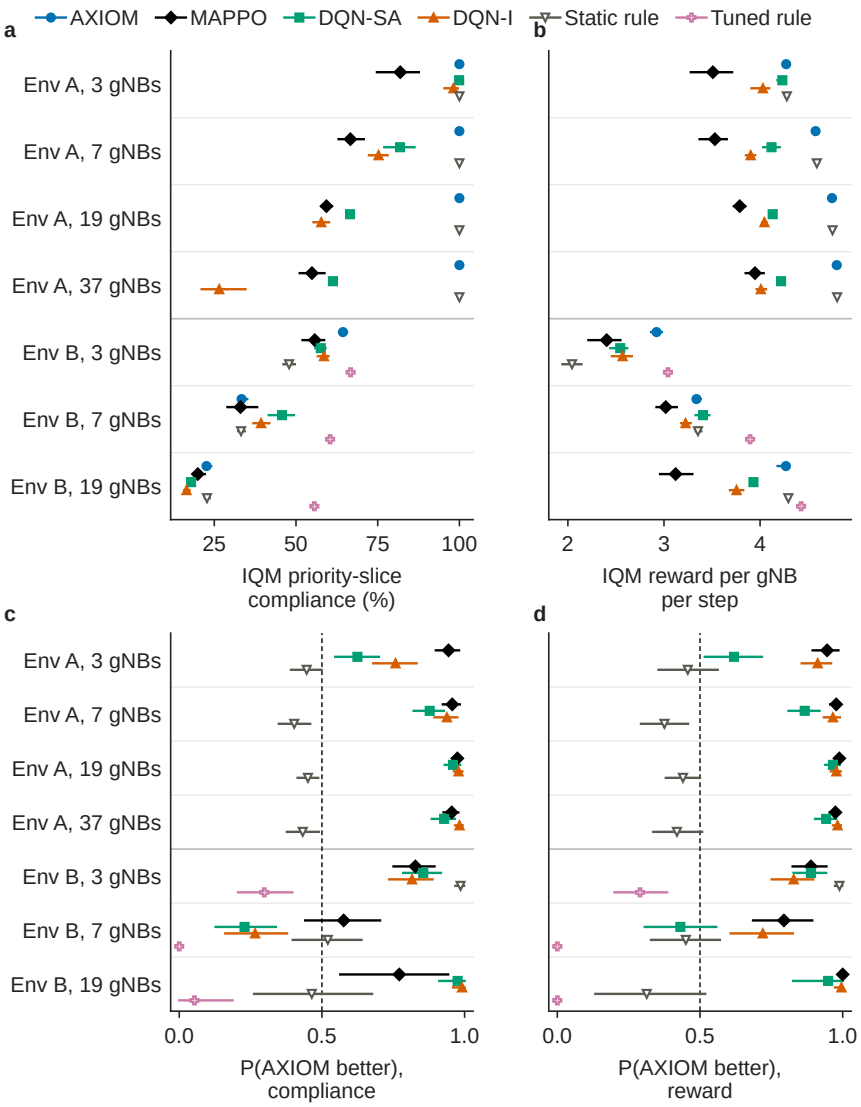


Fig. S3 Interquartile means and probabilities of improvement with stratified bootstrap 95% CIs over 10,000 resamples. a, IQM of priority-slice compliance. b, IQM of reward per gNB per step. c, d, Probability that a run of AXIOM beats a run of each other policy on the same two measures, ties counted as one half, with a dashed line at one half. Online Resource 2 lists every value

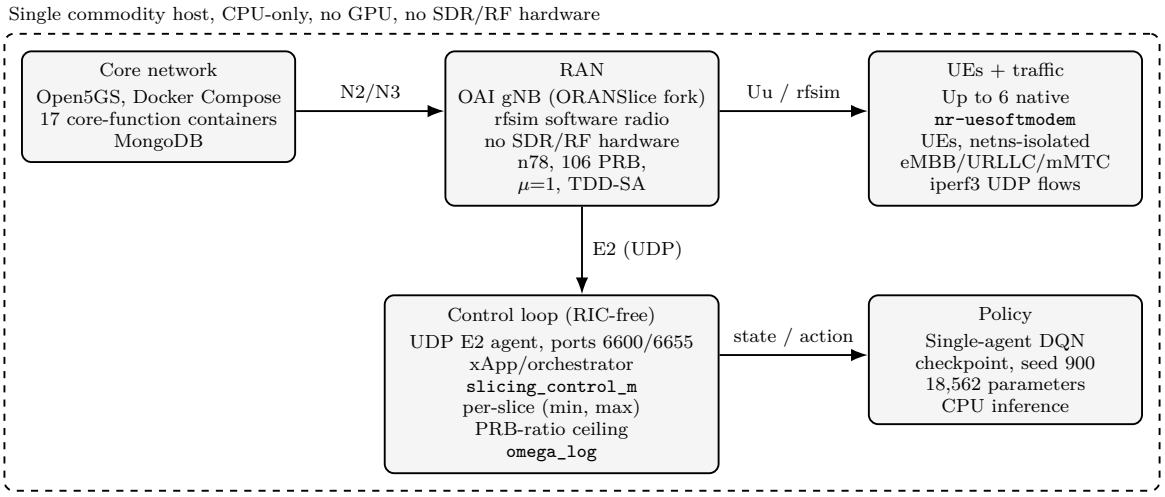


Fig. S4 Real testbed stack, a single-host, software-radio rfsim, RIC-free O-RAN deployment distinguished from software-defined-radio hardware-in-the-loop testbeds. Every block and label is a measured or configured value from this rig

M92, the zero-shot transfer of Section S6, and M94, the distillation of Section S7, were each committed before their runs. M93, the summary of Section S2, reanalyses stored logs only and has no plan of its own. The main paper's Statements and Declarations describe how an artificial intelligence assistant was used.

S5. Three gNBs, ablations, federation, and disruption

The MLP variant reads only a gNB's own features. In Env A AXIOM earns 0.068 more reward per step than the MLP variant at three gNBs, with 95% CI [−0.06, 0.19], a gap that only the paired test on 60 seeds finds significant, at Holm-adjusted $p = 0.015$. It earns 0.31 more at nineteen gNBs on the ring layout, with 7 of 12 seeds tied. A disjoint replication on seeds 1000 to 1009 is stronger for AXIOM. There it earns 1.083, 0.799, and 3.806 more reward per step than DQN-SA, DQN-I, and MAPPO, and pooled over all 40 seeds its margins over the three baselines are 0.504, 1.032, and 2.965 reward per step at Holm-adjusted $p \leq 0.0023$.

A grid of 24 MAPPO configurations raised MAPPO from 60.6% to 69.6% priority-slice compliance at nineteen gNBs on the ring layout, and AXIOM still beats it by 1.25 to 16.18 reward per step at Holm-adjusted $p < 0.001$. With its initialisation properly seeded, DQN-SA keeps 80.3% and 63.9% priority-slice compliance at seven and nineteen gNBs, and AXIOM beats it under every layout at Holm-adjusted $p \leq 0.035$.

Fig. S5b and c show the federated campaign. With federation and no noise, mean reward per step on the ten shared seeds is 12.356 for centralised AXIOM and 11.929 for the federated variant. That is a difference of +0.427 with 95% CI [−0.54, 1.51] and $p = 0.65$, so federation shows no significant cost on these seeds. Each client takes 17,969 noisy steps over its training run. At δ=10^{-5} Eq. (3) then gives ε of about 146,000, 37,000, 9,600, and 2,600 for σ of 0.5, 1, 2, and 4, so a useful guarantee would need noise on the roughly 360 per-round model updates instead.

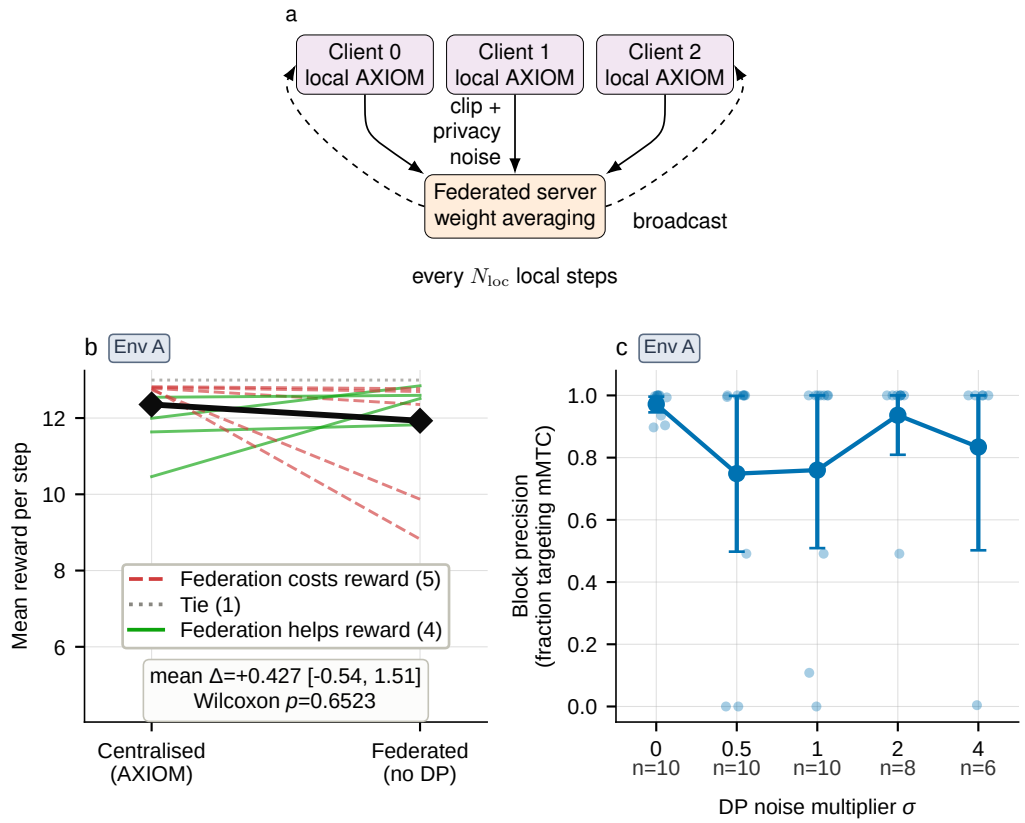


Fig. S5 Federated training in Env A. a, The federated variant, each gNB a client synchronised by periodic federated averaging with optional privacy noise. b, Paired per-seed federation cost on mean reward per step over 10 seeds, not significant at $p = 0.65$. c, Block precision versus DP noise multiplier σ, over the seeds that block

Each policy's frozen Env A checkpoint also faced three disruptions. A gNB dropout forces one gNB to reject everything, a demand spike multiplies the arrival rate, and an agent churn swaps one gNB's decisions for an untrained copy. Dropout and churn windows last 10%, 30%, or 60% of the episode. Spikes multiply arrivals by 2, 4, or 8. Fig. S6 shows the reward each policy keeps, normalised by each step's pending-request count.

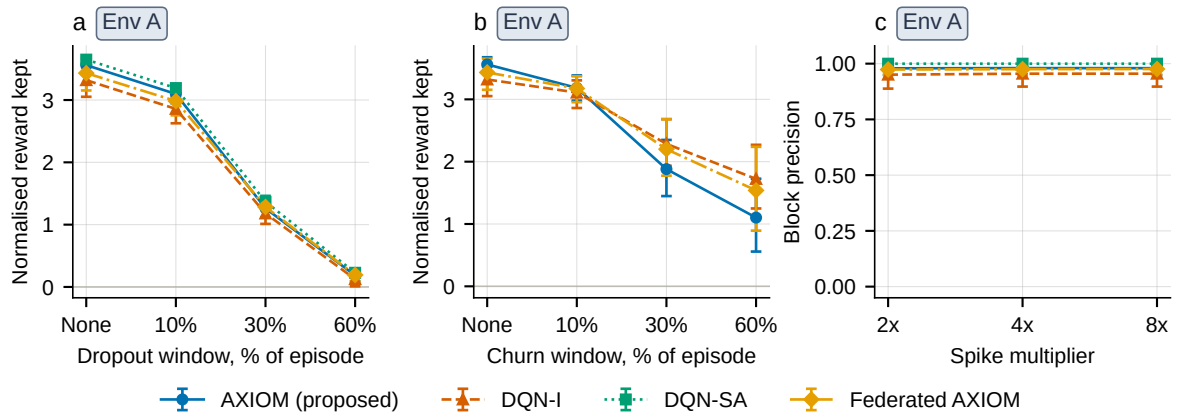


Fig. S6 Disruption campaign, 10 seeds per policy, 95% CIs. a, b, Normalised reward kept under dropout and churn, with the undisrupted level at the tick marked None. c, Block precision under demand spikes

S6. Scaling to thirty-seven gNBs and stronger baselines

In the gate map of the main paper's thirty-seven-gNB cluster, DQN-SA leaves every admission gate open at 0.93 to 0.95 and DQN-I closes the mMTC gate only to 0.95. MAPPO there partly closes the eMBB gate to 0.84 but leaves mMTC open at 0.56. At seven gNBs AXIOM keeps URLLC and eMBB at 99.8% and 94.4%, against 87.6% to 94.9% and 49.3% to 64.6% for the baselines, and at thirty-seven at 97.7% and 92.0%. The thirty-seven-gNB configuration extends the smaller ones, with 37 arrivals per step and at most 148 pending requests, and Table S2 lists every policy. AXIOM's nine collapsed runs keep 57% to 65% priority-slice compliance, while its other runs keep 95.8% to 98.5% on average under each layout.

Table S2 Env A with thirty-seven gNBs, reported and fresh seeds. Priority-slice compliance is the mean of URLLC and eMBB compliance, and collapse counts policies that accept every request

Policy	Runs	URLLC	eMBB	Priority SLA	Reward	Collapsed
AXIOM, hex	60	97.7%	92.1%	94.9%	175.20	5
AXIOM, ring	30	96.1%	88.6%	92.4%	173.20	3
AXIOM, fully-connected	30	99.2%	95.2%	97.2%	176.11	1
DQN-SA	60	72.7%	44.3%	58.5%	144.84	54
DQN-I	60	39.8%	25.7%	32.7%	148.87	7
MAPPO	60	66.9%	45.8%	56.3%	145.12	0
Static rule	60	100.0%	100.0%	99.98%	177.70	0
Fresh seeds 6100 to 6119, hex layout						
AXIOM	20	95.4%	94.7%	95.1%	174.67	1
DQN-SA	20	71.6%	43.1%	57.4%	139.20	18
DQN-I	20	42.0%	26.1%	34.1%	149.04	2
MAPPO	20	70.1%	47.0%	58.5%	144.04	0
Static rule	20	100.0%	100.0%	99.97%	177.68	0

The DQN baselines were also tuned on pilot seeds over learning rates from 10^{-4} to 10^{-3} and minibatches of 16 and 64, as Table S3 lists. No tuned baseline exceeds the accept-all floor at either size, and AXIOM beats each one on compliance and reward in every setting, winning 26 to 30 of 30 seeds at Holm-adjusted $p < 0.001$. VDN, which trains DQN-I's per-gNB networks on one joint target, ties AXIOM at three gNBs but falls behind at nineteen, as Table S4 shows.

Table S3 DQN baselines tuned on pilot seeds, Env A, 30 hex seeds. LR is the learning rate, batch the minibatch size, and Before the priority-slice compliance before tuning

Baseline	gNBs	LR, batch	Priority SLA	Before	Reward	Collapsed
DQN-SA	19	10^{-4}, 16	67.1%	65.3%	78.55	30
DQN-I	19	$3{\times}10^{-4}$, 64	59.1%	56.4%	77.07	11
DQN-SA	37	10^{-3}, 16, unchanged	58.7%	58.7%	144.70	26
DQN-I	37	10^{-4}, 64	45.5%	32.1%	150.42	18

Table S4 VDN against AXIOM and DQN-I, reward per step and priority-slice compliance, 30 seeds per setting

Setting	VDN	AXIOM	DQN-I	VDN SLA	AXIOM SLA	DQN-I SLA
Env A, 3 gNBs	12.664	12.698	11.587	99.0%	98.9%	92.9%
Env B, 3 gNBs	8.552	8.832	7.794	62.5%	65.1%	59.0%
Env A, 19 gNBs, hex	86.428	89.161	76.388	83.4%	97.0%	56.4%

Fig. S7 gives the per-layout view of the Env A scaling results that the main paper pools over layouts.

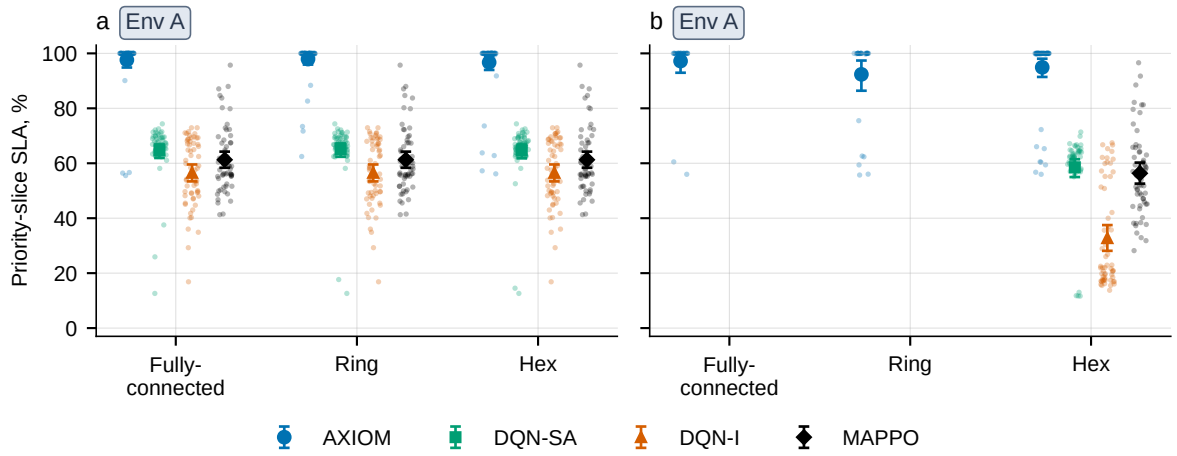


Fig. S7 Env A priority-slice compliance by layout, per-seed points behind the mean and 95% CI. a, Nineteen gNBs, 60 seeds per layout. b, Thirty-seven gNBs, where the baselines never read the layout and ran once, on the hex seeds

The transfer run of the main paper's Section 6.2, M92, also covered AXIOM's MLP variant in Env B. The stored network also holds a mixer left from an earlier training design. No current loss uses it and its input grows with the cluster, so the loader copies every other parameter and AXIOM keeps its 7,782 execution parameters. Before the plan was committed, six stored evaluations were rerun through the same loader at their own size, and each reproduced exactly. Table S5 adds CIs, reward, and collapses to the main paper's transfer table. In Env B the endpoint is per-gNB compliance, because cluster-level compliance, which needs all nineteen gNBs to comply at once, stays between 18% and 36% for every policy there. Here graph attention adds nothing measurable, since AXIOM transferred from seven gNBs keeps 0.80 points more per-gNB compliance than its MLP variant, with 95% CI $[-2.23, 3.81]$ and $p = 0.72$. On the fully-connected layout at nineteen gNBs, AXIOM trained at that size makes exactly the static rule's decisions on every seed, so the transferred policy trails it by 0.10 reward per gNB, at $p = 0.068$.

Table S5 Zero-shot transfer from seven gNBs on 15 paired seeds, reward per gNB per step and collapsed runs. The main paper's transfer table gives priority-slice compliance with 95% CIs. The MLP variant moved from seven gNBs keeps 78.3% per-gNB priority-slice compliance in Env B at nineteen gNBs, with 95% CI [75.5, 80.7]

	AXIOM from 7		AXIOM at the size		MAPPO from 7
Env, layout, gNBs	Reward	Collapsed	Reward	Collapsed	Reward
A, hex, 19	4.645	0	4.659	1	3.989
A, hex, 37	4.690	0	4.733	0	4.134
A, ring, 19	4.666	0	4.689	0	3.989
A, ring, 37	4.711	0	4.689	2	4.134
A, fully connected, 19	4.654	0	4.756	0	3.989
A, fully connected, 37	4.702	0	4.715	1	4.134
B, hex, 19	4.257	0	4.225	1	3.898
MLP variant from 7	4.237	0			

S7. Env B, shifted load, longer training, and convergence

No deployable policy we tested keeps eMBB compliance above 37% in Env B, because demand at the load peaks exceeds the budget by up to 23.5%. Even the rule with the highest eMBB compliance among all 95 local rules, chosen in hindsight on each evaluation stream, reaches only 37.7% on average. Table S6 lists AXIOM against the rule tuned on nominal conditions, on 25 checkpoints neither retrained nor re-tuned. A rule re-tuned on the shifted training streams selected thresholds different from the nominal rule's at every shift. Against it AXIOM trails by 0.16 and 0.12 per step at 110% and 115% load, which is not significant, and by 0.62 at 120% load.

Table S7 lists Env B at nineteen gNBs on the hex layout. Per-gNB compliance counts a slice as compliant at a gNB and step when that gNB meets the slice's SLA, whereas the cluster-level metric counts a violation at any gNB as a violation of the cluster. AXIOM makes exactly the static rule's decisions in 9 of the 15 seeds. In 4 of the other 6 it also blocks only mMTC, one collapses and blocks nothing, and one also blocks other slices. On 15 fresh seeds, 7000 to 7014, in Env B at nineteen gNBs AXIOM kept 76.9% per-gNB priority-slice compliance. It earned 5.90 to 22.00 more reward per step than each learned baseline and kept 6.9 to 11.1 points more per-gNB priority-slice compliance, at Holm-adjusted $p \leq 0.011$.

Table S6 AXIOM against the rule tuned on nominal conditions, Env B with three gNBs, 25 seeds, reward per step, wins over losses, and the unadjusted Wilcoxon p

Condition	AXIOM	Tuned rule	Difference	AXIOM wins	p
80% load	13.776	14.055	−0.279	12/13	0.44
nominal	8.846	9.261	−0.415	7/18	0.00016
105% load	4.482	4.183	+0.299	21/4	0.017
110% load	2.082	1.624	+0.458	23/2	8.8e-05
115% load	0.378	−0.052	+0.431	23/2	1e-05
120% load	−0.714	−1.068	+0.354	21/4	8.8e-05
coupling 0.30	6.457	6.766	−0.310	11/14	0.28
load swing 0.50	6.015	5.831	+0.184	18/7	0.045

Table S7 Env B with nineteen gNBs, 15 seeds, hex layout. SLA columns give priority-slice compliance, the mean of URLLC and eMBB compliance, per gNB and at cluster level

Policy	Reward	Per-gNB SLA	Cluster SLA	Collapsed
AXIOM	80.28	77.3%	24.7%	1
DQN-SA	74.74	66.4%	18.0%	14
DQN-I	71.20	66.8%	16.7%	0
MAPPO	59.07	71.8%	20.8%	0
Static rule	81.59	79.9%	23.0%	0
Accept all	74.77	66.4%	18.0%	15
Tuned local rule, carried over	84.09	84.5%	55.7%	0

At seven gNBs the tuned rules earn 26.5 to 27.3 per step and beat AXIOM in all 15 seeds under every layout, while AXIOM earns 23.4 to 23.6, no more than the static rule's 23.6. The reward charges each slice's SLA penalty once for the cluster when any gNB violates it, so shedding eMBB at one gNB pays only when the other gNBs also comply. Table S8 lists three changes on the ring layout. The per-gNB SLA penalty raised AXIOM's compliance by 10.7 points, a learning rate of 0.003 by 14.9, and 900 training episodes by 21.7. The Holm-adjusted p values are 0.007, 0.007, and 0.0004. With 900 episodes for every learner, AXIOM keeps 12.8 to 15.2 points more compliance and earns 0.44 to 0.82 more reward per gNB than each baseline, at Holm-adjusted $p \leq 0.0012$. The tuned rule still keeps more compliance than AXIOM in all 15 seeds. At nineteen gNBs the longer budget does not help. After 900 episodes seven of eight seeds still shed only mMTC, against all eight after 300. Under uneven load, with each gNB's multiplier drawn between 75% and 125% per seed, AXIOM does not beat the nominally tuned rule at seven gNBs, earning 0.36 more reward per step at $p = 0.23$. It still beats every learned baseline there, by 1.18 to 4.28 reward per step at unadjusted $p \leq 0.005$. Under the per-gNB SLA penalty every learned baseline kept 13.7 to 18.5 percentage points less per-gNB compliance than AXIOM, in every seed.

Table S8 Env B, seven gNBs, ring layout, 15 seeds, reward per gNB per step, priority-slice compliance, and seeds that shed only mMTC

Policy	Reward per gNB	Priority SLA	mMTC-only seeds
AXIOM, 300 episodes, as reported	3.36	34.0%	11
AXIOM, per-gNB SLA penalty	–	44.7%	0
AXIOM, learning rate 0.003	3.50	48.9%	3
AXIOM, 900 episodes	3.79	55.7%	1
DQN-SA, 900 episodes	3.35	41.9%	1
DQN-I, 900 episodes	3.29	42.9%	0
MAPPO, 900 episodes	2.97	40.5%	0
Tuned local rule	3.91	60.5%	0

Fig. S8 shows what AXIOM learned at seven gNBs on the ring layout. After 300 episodes the eMBB head of 11 of its 15 seeds would reject at no step. After 900 episodes 14 of 15 seeds reject more often as the gNB's own URLLC queue grows, a softer form of the tuned rule's step at 0.2.

The distillation of the main paper's Section 6.3, M94, replays each stored checkpoint on its training and held-out streams, and the held-out replays reproduce the stored evaluations exactly. Each request gives four features, the gNB's URLLC queue, its congestion level, the requesting slice's PRB share, and the mean congestion of its neighbours. One tree per slice is fitted on training-stream decisions at the depth that 5-fold cross-validation selects, which was 3 for every target, and Table S9 gives the result. AXIOM rejected no URLLC request in any replay, and 34 of the 60 mMTC trees reject every request. The eMBB trees split first on the URLLC queue in 24 of 30 seeds at three gNBs, at a median of 0.21. At seven gNBs after 900 episodes they do so in 13 of 15 seeds, at a median of 0.28, against the tuned rule's 0.2.

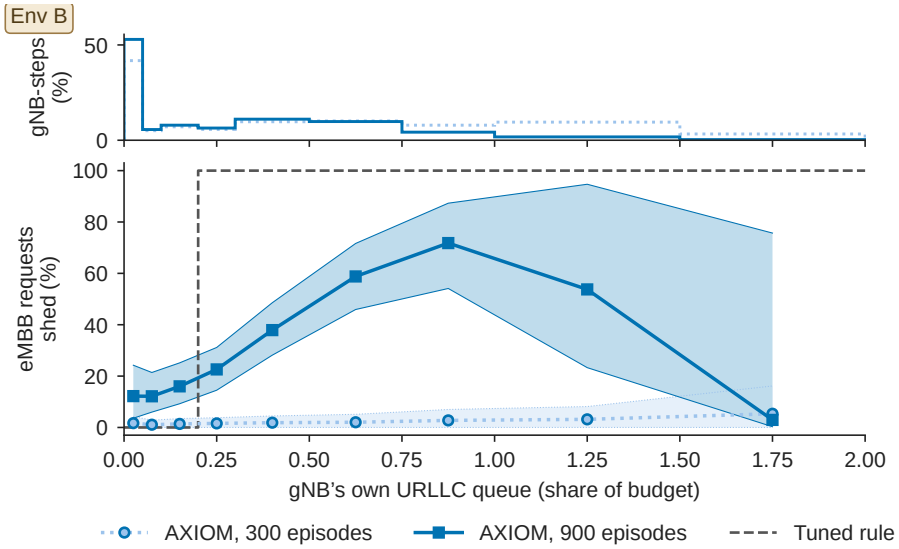


Fig. S8 What AXIOM learned in Env B at seven gNBs on the ring layout, 15 seeds. Share of gNB-steps on which its eMBB head would reject, by the gNB's own URLLC queue, with 95% CIs over seeds, beside the tuned rule's step at 0.2. The marginal above gives how often each queue level occurs

Table S9 AXIOM distilled into one decision tree per slice in Env B. Fidelity is the share of held-out decisions on which the trees agree with AXIOM, by tree depth. The last columns run the depth-3 trees as a rule, with 95% CIs and Holm-adjusted p

		Fidelity by depth			Rule minus AXIOM, reward		Rule minus tuned rule	
Checkpoints	Seeds	1	2	3	Difference	p	Difference	p
3 gNBs, 300 episodes	30	0.833	0.861	0.862	−0.189 [−0.29, −0.09]	0.0016	−0.609 [−0.74, −0.47]	<0.001
7 gNBs ring, 300 episodes	15	0.963	0.964	0.974	0.057 [−0.54, 0.78]	1	−3.783 [−4.61, −2.91]	<0.001
7 gNBs ring, 900 episodes	15	0.927	0.925	0.931	−0.147 [−0.29, 0.00]	0.083	−0.959 [−1.63, −0.40]	0.0026

S8. Throughput and delay

Table S10 lists throughput, the share of offered demand that is served, and delay, the mean backlog over the mean served PRBs in 5 s control periods, with the 95th percentile over steps and gNBs. This delay is a queueing proxy from the fluid model, not packet latency. Each re-run with a stored log reproduced it exactly, and the Env A accept-all rows, which have no stored log, are marked unchecked.

Table S10 Served share of offered demand and backlog delay in control periods, mean and 95th percentile, per slice

Policy	URLLC served	eMBB served	URLLC delay	p95	eMBB delay	p95
Env A, three gNBs						
AXIOM	99.8%	99.7%	0.11	0.66	0.11	0.47
DQN-SA	99.5%	99.4%	0.20	0.96	0.18	0.83
DQN-I	99.4%	98.5%	0.23	1.11	0.43	6.24
MAPPO	98.4%	90.5%	0.38	3.31	3.81	19.67
Static rule	99.8%	99.9%	0.10	0.59	0.05	0.31
Accept all, unchecked	97.5%	97.6%	0.80	3.01	0.76	2.65
Env A, thirty-seven gNBs, hex						
AXIOM	99.8%	99.0%	0.08	0.42	0.34	1.12
DQN-SA	94.2%	95.2%	3.61	16.87	2.09	14.28
DQN-I	97.7%	97.5%	0.76	5.23	0.72	5.49
MAPPO	96.4%	85.7%	0.58	5.38	6.28	35.70
Static rule	100.0%	100.0%	0.03	0.17	0.02	0.09
Accept all, unchecked	98.7%	98.8%	0.43	2.14	0.38	1.83
Env B, three gNBs						
AXIOM	98.4%	88.5%	0.65	2.32	3.18	19.28
DQN-SA	98.2%	82.1%	0.75	3.15	5.73	49.79
DQN-I	98.6%	82.7%	0.66	2.81	4.73	49.34
MAPPO	97.2%	83.5%	1.10	4.00	4.57	43.82
Static rule	96.0%	96.1%	1.59	5.25	1.57	5.11
Tuned rule	99.0%	84.8%	0.58	1.96	3.57	38.55
Accept all	92.8%	92.9%	2.59	7.72	2.58	7.59
Env B, nineteen gNBs, hex						
AXIOM	96.6%	92.6%	1.35	4.67	3.41	16.01
DQN-SA	93.8%	93.9%	2.26	6.99	2.24	6.82
DQN-I	94.1%	86.1%	2.17	9.71	4.17	72.77
MAPPO	89.2%	88.3%	2.66	33.84	2.58	28.13
Static rule	96.8%	96.9%	1.33	4.65	1.32	4.50
Tuned rule, carried	99.1%	87.9%	0.51	1.77	2.78	25.40
Accept all	93.8%	93.9%	2.25	6.97	2.24	6.82

S9. Reproducing the results

Table S11 lists what reproducing the results needs and the order of the steps. Every result in both documents comes from one laptop without a graphics processor, and every simulation run fixes its seed.

Table S11 Reproducing the results. The hardware and software every run used, and the steps from the stored code to the figures and the manuscript. The live testbed is described in Section S3

Item	Details
What is needed	
Hardware	One laptop, Intel Core i5-1135G7, 4 cores and 8 threads. 8 GB of memory, 7.4 GiB usable, with 8 GiB of swap. No graphics processor, every run on the CPU. Ubuntu 24.04 LTS
Software	Python 3.12.3, PyTorch 2.13.0 on the CPU. NumPy 2.5.1, SciPy 1.18.0, Matplotlib 3.11.0, PyYAML 6.0.3. TeX Live 2023 with pdfLaTeX. Simulators and learners in `framework/`, never modified
Live testbed check only	OAI gNB, ORANSlice fork on OAI 2024.w28, rfsim radio, band n78, 106 PRBs. Open5GS core in Docker, 17 containers. protobuf-c 1.5.2, built from source
Steps	
1. Set up	Request the code, configurations, and stored logs from the corresponding author, and create a Python virtual environment with the packages above. Every job runs from `framework/` on one thread, with `OMP_NUM_THREADS=1`, `MKL_NUM_THREADS=1`, and `OPENBLAS_NUM_THREADS=1`.
2. Train and evaluate	One command trains and evaluates one policy on a batch of seeds, for example `M63_BACKLOG_RESET=1 M64_AXIOM_V2=1 python m64_run.py --arm axiom2 --env A --n 19 --topology ring --seeds 900 901`. Each seed trains for 300 episodes on its own stream and is then evaluated for 50 episodes on the held-out stream 5000 plus the seed. The median training time per seed runs from 73 s to 2,088 s, and the main campaigns ran five jobs at a time and the seed extension six. The primary seeds are 900 to 929, the replication 1000 to 1009, the pilots 2000 to 2009, the fresh seeds 3000 onwards, and the seed extension 11000 onwards.
3. Run the other campaigns	`m3_run_experiment.py` trains the federated variant with noise multipliers 0, 0.5, 1, 2, and 4, `m4_run_experiment.py` applies the disruptions, `m73_run.py` adds 37 gNBs, `m70_run.py` to `m97` add the follow-up runs, and `m107_run.py` adds the seed extension.
4. Re-evaluate stored checkpoints	`m72_shift.py` replays AXIOM and the tuned rule under shifted conditions, `m77_kpi.py` records served PRBs and backlog, `m92_transfer.py` runs checkpoints at a new size, and `m94_distil.py` distils AXIOM into trees.
5. Summarise	Each campaign's `analysis` script judges every pre-registered criterion in `docs/PAPER5_M*_preregistration.md` and reports whether it holds or fails.
6. Verify	`m66_verify.py` recomputes every results table and key sentence of both documents from the raw logs with its own reader and Holm correction, `m82_audit.py` traces the remaining numbers to configurations and code, and `m66_ci_recompute.py` recomputes every bootstrap CI.
7. Redraw and build	`paper5_regen_wpc_figures.sh` redraws every plot, `paper5_cvd_check.py` checks each for colour vision, and `make_submission.py` writes `Fig1` to `Fig10`, `ESM_1.pdf`, `ESM_2.csv`, and the compiled manuscript.